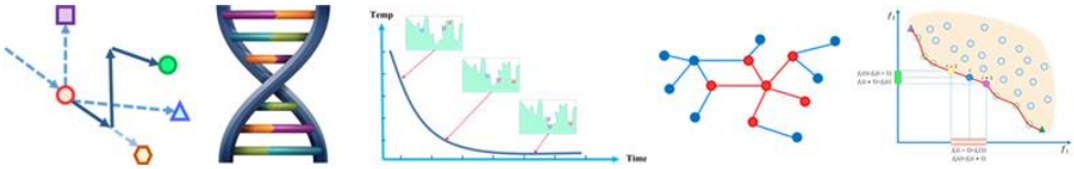
OR 6500: Metaheuristics and Applications



Prof. Mohammad Dehghani

"Evolution continually innovates, but at each level it conserves the elements that are recombined to yield the innovations."
- John Henry Holland - Developer of Genetic Algorithm

Metaheuristics and Applications

OR6500

Instructor Info

Mohammad Dehghani

Office Hrs: Tuesdays 3:00pm-4:30pm

Zoom (Make an appointment)

Website

m.dehghani@northeastern.edu

Course Info

Prerequisite:
* OR 6205: Deterministic OR
* Basic knowledge of MATLAB

Bi-weekly Sessions Mondays

2:30pm-3:15pm

TA Info

Sahil Belsare

Office Hrs: TBD

Virtul Meeting & On ground

belsare.s@northeastern.edu

Course Overview

Metaheuristic search algorithms is a class of algorithms that can be used to solve large combinatorial optimization problems. In these problems it is usually impossible to find the globally optimal solution on a fast computer, therefore the metaheuristic search aims to find a 'good' solution that satisfies the problem constraints. This course describes multiple metaheuristic search methods including Simulated Annealing (SA), Tabu Search (TS), Genetic Algorithms (GA), Particle Swarm Optimization (PSO), and multi-objective methods. Algorithms will be used to find values of discrete and/or continuous variables that optimize system performance. The application of metaheuristics to a variety of different problems including, Hub Location Allocation, Parallel Machine Scheduling, Travelling Salesman Problem (TSP), Curve Fitting, Clustering, nQueen, Min One, etc., will be discussed. Some practical experiments to demonstrate the advantages and disadvantages of metaheuristic search methods for different applications will be carried out as well.

Course Objectives

- Understand the concepts of metaheuristic methods and how they are used as a tool to solve global optimization problems
- Be able to implement the algorithms in a programming environment, and handle integer and/or continuous variables without any major extension in the same framework
- Apply metaheuristic algorithms in order to solve engineering problems such as: scheduling of a manufacturing system or an emergency room (ER), hub location allocation, data analytics, transportation, traveling salesman problem (TSP), etc.
- Understand the concept of multi-objective optimization and its metaheuristic methods
- Be able to improve the robustness of algorithms by performing experimental analysis and setting the algorithm parameters

Textbook & Software

- Main Textbook
  - Essentials of Metaheuristics, Sean Luke, Online Version 2.2, 2015. (Link)
  - NOTE: This book is required for the class (either the hard-copy or its PDF version). In addition, several articles will be provided as the metaheuristic methods are presented in the class.
- Software Packages
  - MATLAB: Please refer to the course setup file for instruction on how to download/get access to MATLAB.

Outline

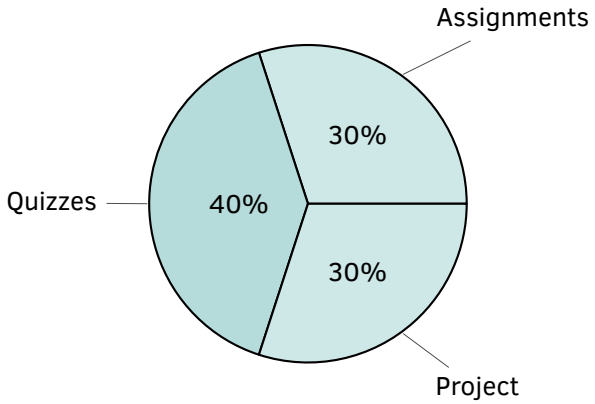
The following are important topics of the course:

- Introduction of Metaheuristics
- Single-objective algorithms
  - Simulated Annealing (SA)
  - Genetic Algorithm (GA)
  - Particle Swarm Optimization (PSO)
  - Tabu Search (TS)
- Multi-objective algorithms
  - Multi-objective Optimization Concept
  - Non-dominated Sorting Genetic Algorithm II (NSGA-II)
  - Multi-objective PSO (MPSO)

Class Schedule

Please check Canvas for the weekly schedule of the course. If due to unforeseen circumstances the schedule changes, students will be informed prior to the change in schedule.

Grading Scheme



- Assignments (Individual)
  - Student are asked to customize or extend the algorithms for different modifications and needs, as per the problem statement.
  - Assignments will be released on a weekly basis.
  - All homework shall be submitted through appropriate mediums as communicated by the professor/TAs
  - If due to unforeseen circumstances the schedule changes, students will be informed prior to the change in schedule. Submissions have to be labeled as "OR 6500 - HW # (or Lab #) - Sec 2 - <Student Name>"
- Quizzes (Individual)
  - Since the concepts in this course are interrelated and often build off each other, you will be assessed on your cumulative understanding of course topics. The quizzes will be posted online through CANVAS and you have just one attempt to take each quiz.
  - The worst quiz will be dropped in the end
  - Once a quiz is posted, you have a few days (usually 5 days) to take it. However, the quiz duration is limited and you have to save and submit your answers within the given time. Since the quizzed are online, the missed quizzed will not be made up.
- Project (Group based)
  - Project provides you an opportunity to apply core principles of Metaheuristics to solve you preferred complex and large combinatorial optimization problem. Also, you are allowed to use your desired algorithm to solve the problem. The details about the project will be announced.
  - The project can be done individually or in a group of 2 students.
  - Please refer to the suggested project timeline to make sure that you have sufficient time or experimentation and analysis.
  - Project includes two important phases:- *(1) Proposal Presentation* and *(2) Final Presentation*.
  - The project requirements and deliverable will be posted on CANVAS.
- Class Participation & Discussion (Highly Recommended)
  - The course is synchronous and attendance is mandatory for students taking the course online or on-ground. To have an effective learning experience, students are encouraged to have continuous interaction with the faculty. A discussion question will be given to each section/break-out rooms to stimulate critical thinking. All students need to actively participate in discussions to articulate their ideas and share their thoughts.
  - Performance will be evaluated based on their participation and quality of response.

Grading

Your grades will be posted on CANVAS. You can check your running grade at any time, so there are no surprises. In fact, you should audit the correct entry of your grades. No grade corrections will be made after the last week of the semester. Final Grades and Course Letter Grades are not curved and are assigned as follows:

Letter Grade	Points	Letter Grade	Points	Letter Grade	Points
A	93-100	B	83-86	C	73-76
A-	90-92	B-	80-82	C-	70-72
B+	87-89	C+	77-79	F	<70

Emailing Us?

Please use the following format for the subject line of your email to avoid miss communication. This subject line will reduce the confusion.

OR6500 - Sec # - Subject Line

Make-up Policy

- No makeup opportunity will be given for any homeworks/lab assignment/projects.
- Students unable to attend classes are responsible for obtaining information regarding announcements and lecture content

Class Policy

- Students are expected to attend all class sessions and are responsible for all materials presented in class. Past experience has shown students who attend regularly and interact with instructors/classmates tend to be better overall.
- Laptops: You may need to use your laptop during the class upon the instructor's notes
- Phones/tablets/smart watches: Electronic devices may not be used during the class. If you have class work, social networking or sports-score browsing, that is more important to do, then please skip class. Life is full of such choices.
- Discussion between students for homeworks and projects is permissible. However the ideas / solutions / codes must not be copied. Each individual submission must be an outcome of individual effort

Recording of Classes

Classes will be recorded to enable all students to review material covered in synchronous classes. Please contact me if you have any concerns.

Academic Integrity

A commitment to the principles of academic integrity is essential to the mission of Northeastern University. The promotion of independent and original scholarship ensures that students derive the most from their educational experience and their pursuit of knowledge. Academic dishonesty violates the most fundamental values of an intellectual community and undermines the achievements of the entire University.

As members of the academic community, students must become familiar with their rights and responsibilities. In each course, they are responsible for knowing the requirements and restrictions regarding research and writing, examinations of whatever kind, collaborative work, the use of study aids, the appropriateness of assistance, and other issues. Students are responsible for learning the conventions of documentation and acknowledgment of sources in their fields. Northeastern University expects students to complete all examinations, tests, papers, creative projects, and assignments of any kind according to the highest ethical standards, as set forth either explicitly or implicitly in this Code or by the direction of instructors.

Go to Academic Integrity website to access the full academic integrity policy.

Student accommodations

Northeastern University and the Disability Resource Center (DRC) are committed to providing disability services that enable students who qualify under Section 504 of the Rehabilitation Act and the Americans with Disabilities Act Amendments Act (ADAAA) to participate fully in the activities of the university. To receive accommodations through the DRC, students must provide appropriate documentation that demonstrates a current substantially limiting disability.

For more information, visit Disability Resource Center website

Northeastern University Copyright Statement

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Diversity and Inclusivity Statement

Northeastern University is committed to equal opportunity, affirmative action, diversity and social justice while building a climate of inclusion on and beyond campus. In the classroom, members of the University community work to cultivate an inclusive environment that denounces discrimination through innovation, collaboration and an awareness of global perspectives on social justice. It is my intention that students from all backgrounds and perspectives will be well served by this course, and

that the diversity that students bring to this class will be viewed as an asset. I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, socioeconomic background, family education level, ability - and other visible and nonvisible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. Your suggestions are encouraged and appreciated.

Please visit the website for complete information on Diversity and Inclusion.

TITLE IX

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance.
Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including male, female, transgender students, faculty and staff.
In case of an emergency, please call 911.

Please visit the website for a complete list of reporting options and resources both on- and off-campus

Hybrid NUflex

Hybrid NUflex allows some portion of the students in each class to be remote and others to be in class. If you have the opportunity to attend class in-person, please come as often as itâs possible for you to do so. Remember your commitment to your faculty and classmates:

- Use the dynamic scheduling tool to set your preferences for in-person or remote attendance. Each week between Monday and Wednesday, youâll select whether you prefer to attend your class sessions remotely or in-person for the following week. You will be able to choose a preference for each individual class session. For example, you can request to come one of your courses in-person on Monday and Wednesday and remotely on Friday.
- Follow through when you commit to attend class in-person. Space in the classroom is limited, and your fellow students and faculty want to see you.
- If you change your mind about in-person class attendance, update your preferences in the dynamic scheduling tool so someone else can take your seat. The dynamic scheduling tool creates a waitlist for students who would like to attend class in-person but received a remote assignment. Beginning in late January or early February, if youâre assigned an in-person seat in a class session for the following week, you have until Saturday at 11:59pm ET to switch to remote for that class session. Switching from in-person to remote frees up a seat for your fellow students on the waitlist to come to class."